



**Mondragon
Unibertsitatea**

**Escuela Politécnica
Superior**

Reinforcement Learning -RL-

Introduction

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This presentation is based on:

Richard S. Sutton and Andrew G. Barto. Reinforcement learning: An introduction.
MIT Press, Cambridge, MA, 2018.

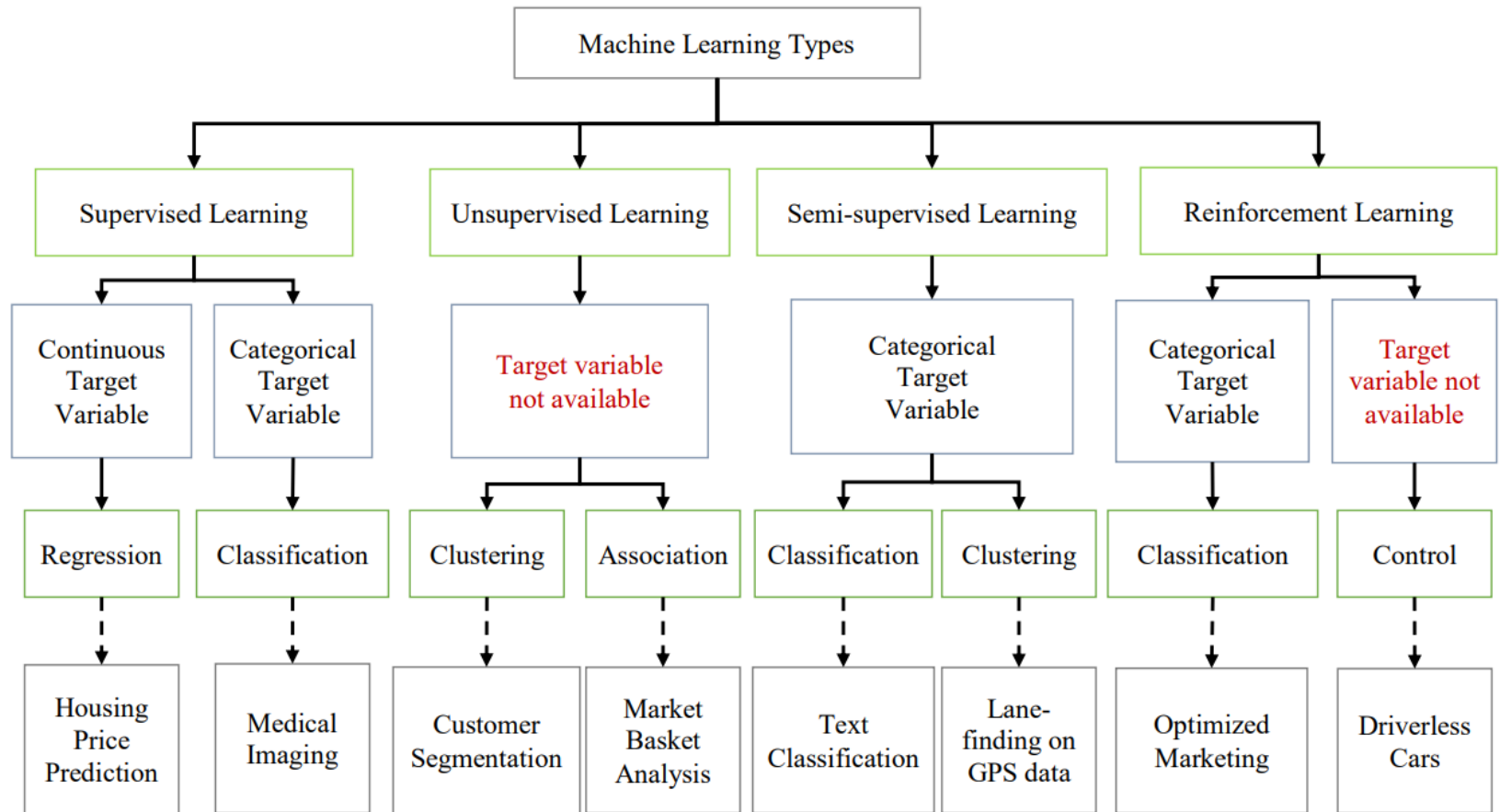
Introduction

- **Course Outline**

1. Introduction
 - General concepts
 - Gymnasium
2. RL Basics
 - Markov Decision Process
3. Classic RL
 - Temporal difference Learning
 - » Sarsa
 - » Q learning
4. Deep RL
 - DQN
 - PPO
5. Evaluation
 - Hyperparameter optimization
 - Evaluation metrics
 - Statistical significance tests

Introduction

- Machine Learning taxonomy**



Shyam, Radhey & Singh, Ria. (2021). A Taxonomy of Machine Learning Techniques. 8. 18-25.

Introduction

- Throughout our lives through interactions we gain knowledge about our environment ourselves.

Supervised Machine Learning

When you try to warn a kid to not touch fire because it will hurt!



Reinforcement Machine Learning

When the kid tries to touch the fire anyway and *learns* that it is not good to touch fire!



<https://thinkingneuron.com/data-science-interview-questions-reinforcement-ml/>

Introduction

Supervised Learning

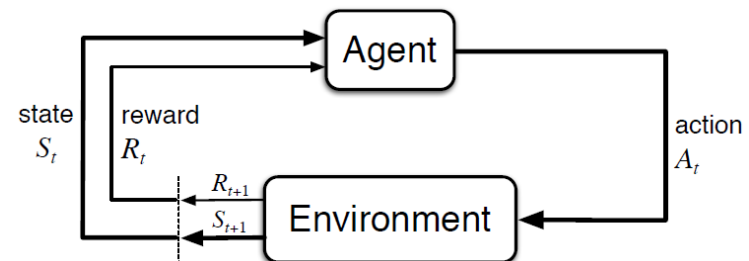
Data Distribution: i.i.d. Assumption Holds

- Data points (X_i, y_i) are Independent and Identically Distributed.
 - Sampling a row from a database does not alter the remaining rows.
- **Feedback Loop: Immediate & Dense**
 - Every prediction $\hat{y}_i$ receives an immediate evaluation via a loss function $\mathcal{L}(\hat{y}_i, y_i)$ against a fixed ground-truth label.

Reinforcement Learning

Data Distribution: i.i.d. is Violated (Sequential Dependence)

- Data is a trajectory
 $(\tau): s_0, a_0, r_1, s_1, a_1, r_2$
- Current data depends heavily on past data.
- The agent's action a_t directly forces a shift in the future data distribution:



Introduction

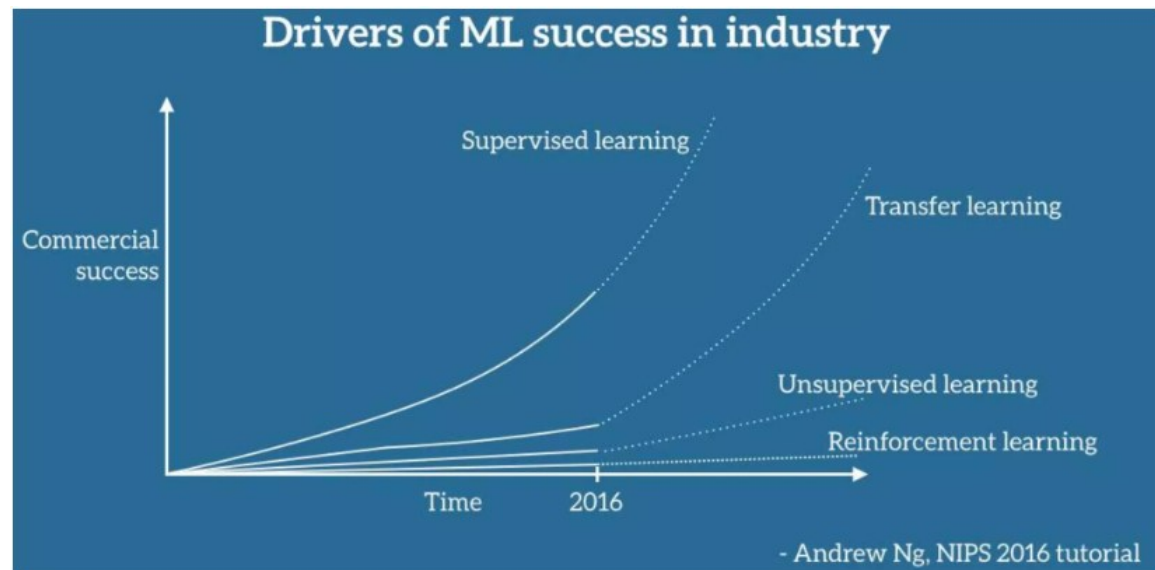
- **What distinguishes reinforcement learning from other machine learning methods?**
 - There is no supervision, only an indicator of reward, how good or bad the action taken has been.
 - Reward is delayed, not instantaneous.
 - The action taken at time t affects future rewards.

Introduction

- RL algorithms typically require millions of interactions to learn effective policies.
- While simulations can help with training, transferring policies from simulation to reality is challenging.
- Designing appropriate reward functions for industrial tasks is complex.
- Industries need consistent, predictable performance.
- Limited interpretability.

But be optimistic. We will see a gradual improvement next years.

- Applications where occasional failures are acceptable.
- Tasks where traditional approaches have failed.



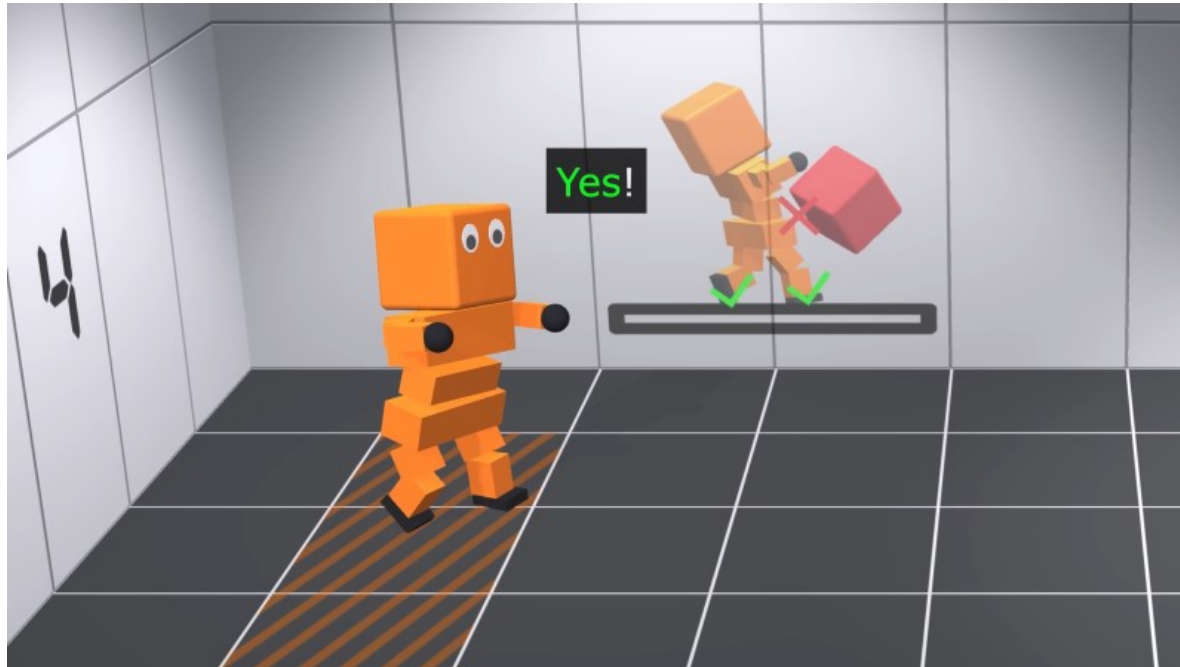
Introduction

- **Examples Reward or “RewArt”**
 - Autonomous driving:
 - +R reward for staying in the lane and maintaining a safe speed
 - -R reward for hitting obstacles or violating traffic rules
 - Robot arm in a factory setting
 - +R reward for successfully completing tasks (e.g., picking, placing objects)
 - -R reward for damaging objects or making errors in assembly
 - Customer service chatbot
 - +R reward for resolving customer inquiries efficiently and with high satisfaction
 - -R reward for misunderstandings or unsatisfactory responses

Introduction

- **Examples:**

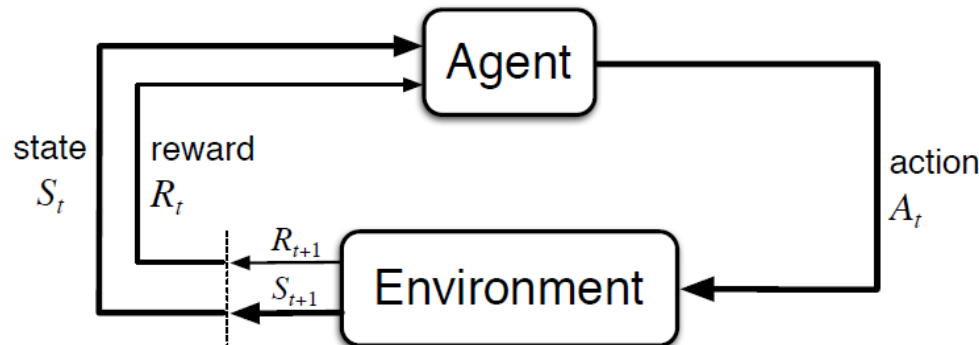
- https://www.youtube.com/watch?v=L_4BPjLBF4E



Introduction

- **The basics**

- **Agent:** interacts over time with its environment to achieve a goal.
- **Environment:** the dynamic landscape where an AI agent interacts and learns
- **Reward:** the agent is rewarded for taking controls that lead to successful states
- **State:** is a representation of the current environment that the agent is in
- **Action:** is a complete description of the state of the world



Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.

Introduction

- **The basics**
 - **Policy:** defines the learning agents way of behaving at a given time. Is a mapping from perceived states of the environment to actions to be taken when in those states.
 - Deterministic policy: $a = \pi(s)$; maps each state to a single action with certainty.
 - Stochastic policy: $\pi(a|s) = \mathbb{P}[A_t = a \mid S_t = s]$; maps each state to a probability distribution over actions
 - **Value function:** Whereas the reward signal indicates what is good in an immediate sense, a value function specifies what is good in the long run. Roughly speaking, the value of a state is the total amount of reward an agent can expect to accumulate over the future.
 - Used to evaluate the goodness/badness of states, and therefore to select between actions

Introduction

- **The basics**

- **Model:** the model refers to a system or framework that simulates the environment's dynamics. It is used to predict the next state and the reward based on the current state and the chosen action.
 - **$\mathcal{P}$ Transition model / Dynamics model:** This predicts the next state s' given the current state s and action a . Describes the probability of transitioning to state s' when taking action a from state s .

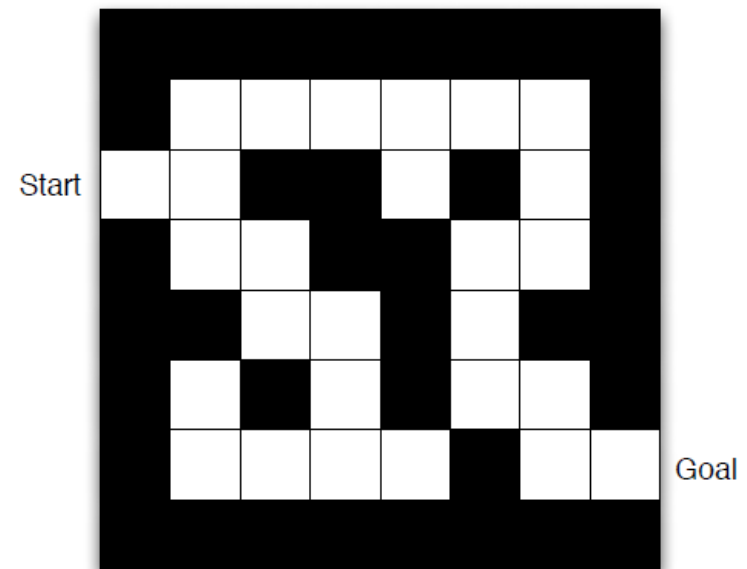
$$\mathcal{P}_{ss'}^a = \mathbb{P}[S_{t+1} = s' | S_t = s, A_t = a]$$

- **$\mathcal{R}$ Reward model:** This predicts the reward r the agent will receive for taking action a in state s . Gives the expected immediate reward for taking action a in state s .

$$\mathcal{R}_s^a = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$$

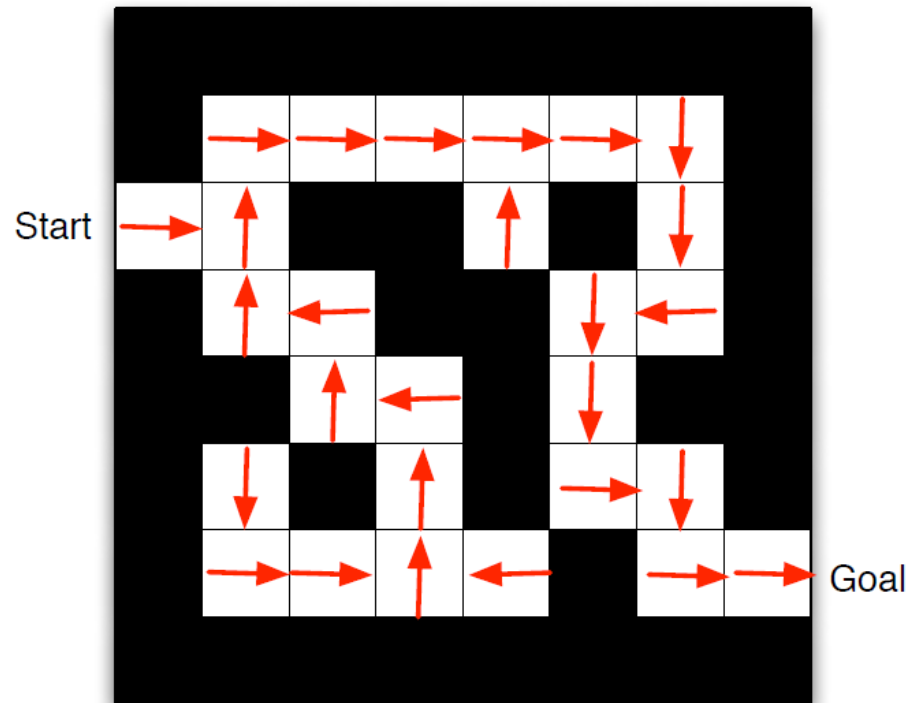
Introduction

- **Maze example**
 - Rewards: -1 per time step
 - Actions: N, E, S, W
 - States: agent location



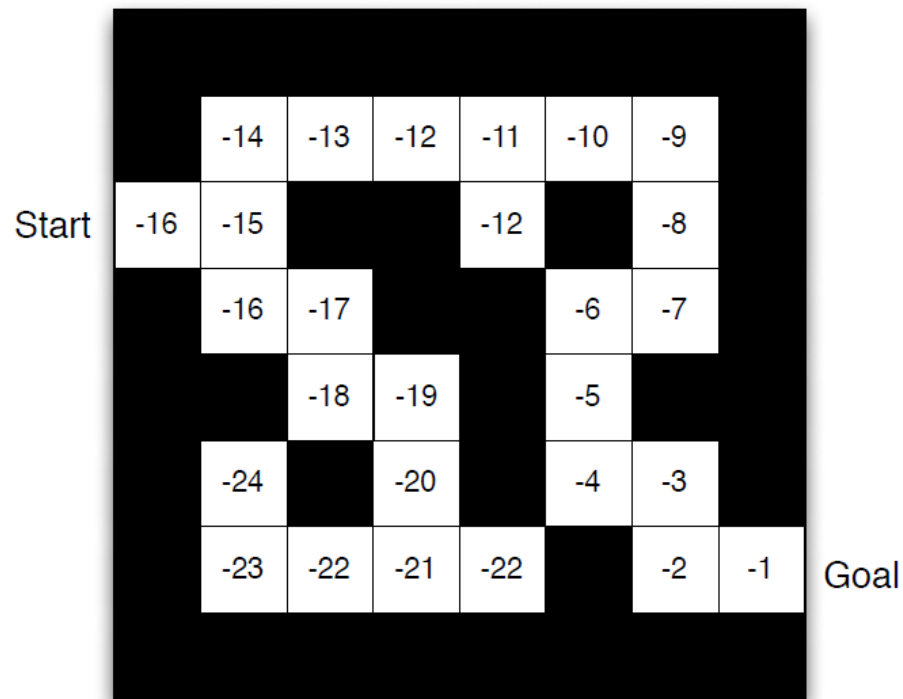
Introduction

- **Maze example: policy**
 - Arrows represent policy $\pi(s)$ for each state s



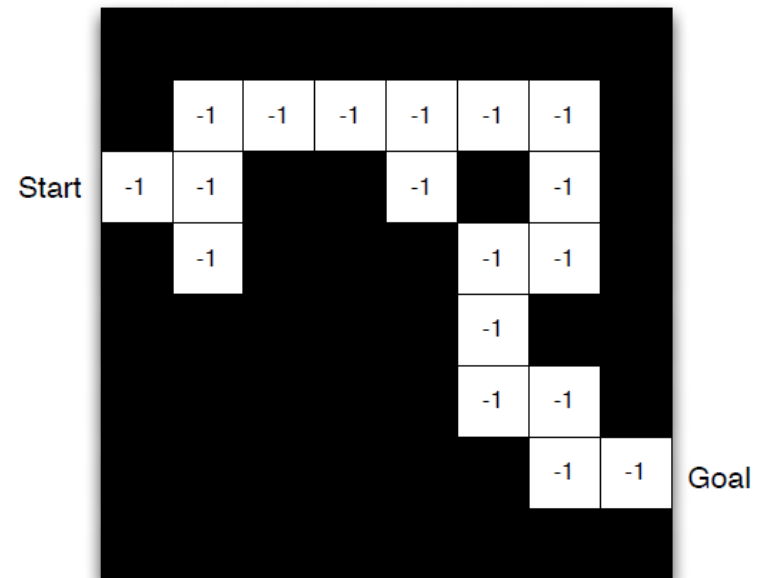
Introduction

- **Maze example: Value function**
 - Numbers represent value $v_{\pi}(s)$ of each state s



Introduction

- **Maze model**
 - Agent may have an internal model of the environment
 - Dynamics: how actions change the state
 - Rewards: how much reward from each state
 - The model may be imperfect
 - Grid layout represent transition model $\mathcal{P}_{ss'}^a$
 - Numbers represent immediate reward $\mathcal{R}_s^a$ from each state s (same for all a)



Introduction

- **Exploration vs exploitation**
 - Reinforcement learning is trial and error
 - The agent should discover a good policy from its experiences of the environment
 - **Exploration:** finds more information about the environment
 - **Exploitation:** exploits known information to maximize reward

It is important to explore as well as exploit

Introduction

- Examples
 - **Self-Driving Cars**
 - A self-driving car must decide on the best route to its destination.
 - **Exploration:** The car takes a new, less-used route to learn if it's faster or has less traffic.
 - **Exploitation:** The car consistently chooses the route it already knows to be the fastest based on historical data.

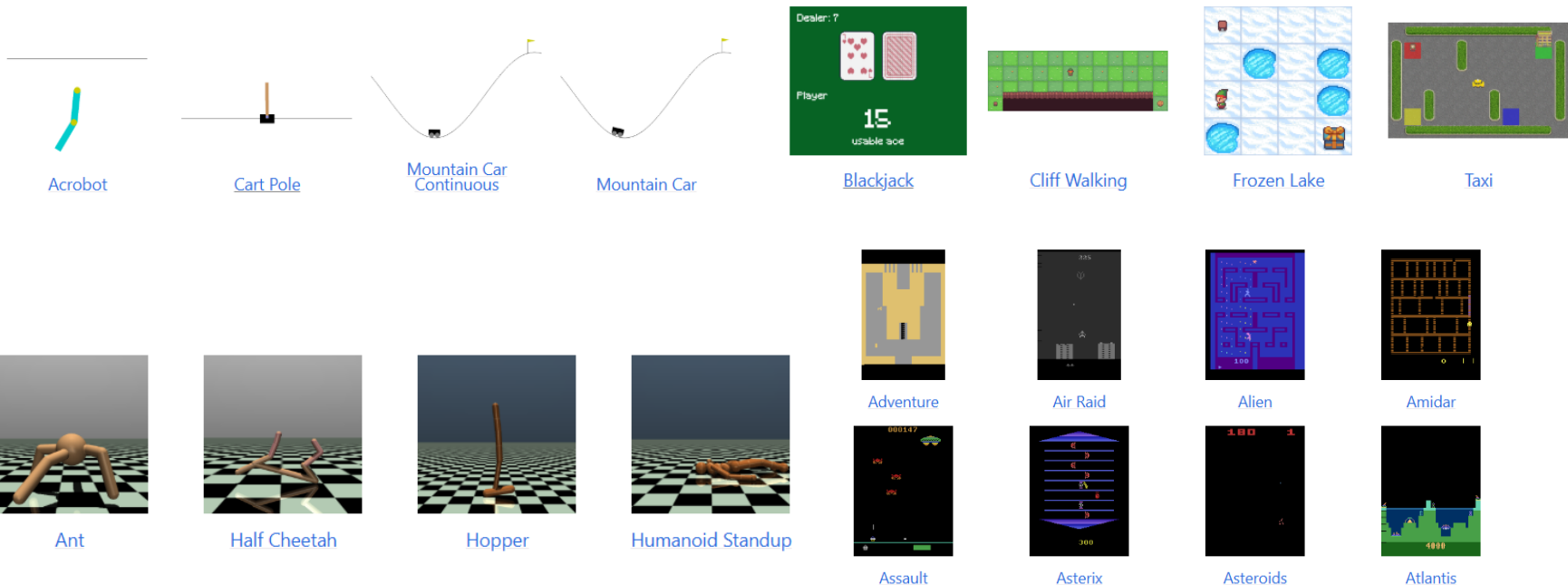
Introduction

- **Key aspects:**
 - The focus is on learning by interacting with an environment, like an opponent player.
 - The existence of a clearly defined objective is a prerequisite for the planning of correct behavior. Such planning must take into account the delayed effects of one's choices.
 - In contrast, tic-tac-toe has a relatively small, finite state set, whereas reinforcement learning can be used in scenarios where the state set is very large, or even infinite.

Introduction

• Gymnasium

- An API standard for reinforcement learning with a diverse collection of reference environments.
- `pip install gymnasium`
- <https://gymnasium.farama.org/>



Introduction

- **Frozen Lake**

- Frozen lake involves crossing a frozen lake from start to goal without falling into any holes by walking over the frozen lake.
- The game starts with the player at location [0,0]
- The player makes moves until they reach the goal or fall in a hole.

- **Action space:**

- 0: Move left
- 1: Move down
- 2: Move right
- 3: Move up

- **Rewards:**

- Reach goal: +1
- Reach hole: 0
- Reach frozen: 0

- **Episode End:**

- **Termination:**

- The player moves into a hole.
- The player reaches the goal at $\max(\text{nrow}) * \max(\text{ncol}) - 1$ (location $[\max(\text{nrow})-1, \max(\text{ncol})-1]$).

- **Truncation (when using the time_limit wrapper):**

- The length of the episode is 100 for 4x4 environment, 200 for FrozenLake8x8-v1 environment.



https://gymnasium.farama.org/environments/toy_text/frozen_lake/

Introduction

- **Exercise:**

- Look to gymnasium documentation and initialize the **FrozenLake-v1** environment and implement an agent that takes random actions until reaching a terminal state for 10 episodes/times.

```
import gymnasium as gym

# Initialise the environment
env = gym.make("FrozenLake-v1", render_mode="human")    render_mode = "rgb_array"

# Reset the environment to generate the first observation
observation, info = env.reset(seed=42)

for _ in range(1000):
    # this is where you would insert your policy
    action = env.action_space.sample()

    # step (transition) through the environment with the action
    # receiving the next observation, reward and if the episode has terminated or truncated
    observation, reward, terminated, truncated, info = env.step(action)

    # If the episode has ended then we can reset to start a new episode
    if terminated or truncated:
        observation, info = env.reset()

env.close()
```

Lunar Lander simple example



Introduction

- General advice

Be patient and persistent

- Results may take **days to train**, and debugging RL models is harder than standard ML models.
- Focus on understanding **why** things work.

Once you see your agent learning and improving, its so satisfying!



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Eskerrik asko
Muchas gracias
Thank you

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